Design Goals and Milestones for PM12 Cooling

Design Laboratory | Planning milestone due 8 November 2026 | Updated 11 November 2026

Problem definition

The ALICE FIT PM12 board needs airflow over the FPGA and three THS788 timing chips. Temperature measurements at all four heatsinks will show how effectively the prototype cools each location.

Operating environment

Bench tests will use a laboratory 12 V supply for the fans and USB power for the Pico, with a common ground. The bracket must leave room for cables and access to PM12 connectors.

Design goals

Independent cooling. Use one Raspberry Pi Pico 2 W to control three 12 V Arctic P9 PWM PST fans independently and measure their RPM via TACH. Read four NTC thermistors through an ADS1115 over I^2C, one sensor per heatsink.

Cooling after shutdown. Keep the Pico and all three fans powered for 60 seconds after PM12 shutdown. Test with a switch before connecting the controller to a verified PM12 status signal.

Monitoring and control. Display temperatures, RPM, PWM settings and the shutdown countdown in the app. Use Wi-Fi for monitoring and manual adjustments. The Pico continues automatic cooling when the app disconnects.

Removable mounting. Mount the fans along the short upper edge on a removable 3D-printed bracket. Direct airflow along the board and across the heatsinks.

Project milestones

Stage	Completion criterion
1 Concept and parts selection	Completed by 11 November. The concept, wiring diagram, parts list and cost estimate are documented.
2 Single-channel bench test	Run one fan and read one NTC through the ADS1115. Verify PWM control, RPM measurement and response to a temperature change.
3 Three-fan prototype	Verify independent PWM and RPM readings for three fans and temperature readings from four NTCs. Test the 60-second run-on with the switch.
4 Mounting and monitoring	Fit the printed bracket and verify live readings and independent fan commands in the app over Wi-Fi.
5 Validation and demonstration	Compare heatsink temperatures with fans off and on under the same workload. Check the farthest heatsink, record fan noise and verify run-on after PM12 shutdown.